LEARNING MODULE

6

Industry and Competitive Analysis

LEARNING OUTCOMES

Mastery	The candidate should be able to:
☐	describe the purposes of, and steps involved in, industry and competitive analysis
☐	describe industry classification methods and compare methods by which companies can be grouped
☐	determine an industry's size, growth characteristics, profitability, and market share trends
☐	analyze an industry's structure and external influences using Porter's Five Forces and PESTLE frameworks
☐	evaluate the competitive strategy and position of a company

INTRODUCTION

1

It is essential for analysts to understand the industry context for a company's financial performance and its evaluation. For example, is a company's revenue growth driven by a macroeconomic or industry-wide factor (such as an economic expansion) or a company-specific factor (such as increasing market share)? The answer has important implications for forecasting and valuation.

In this learning module, we discuss how an industry is defined and address the challenges associated with grouping companies that operate in multiple industries. Next, we introduce methods to survey an industry in terms of size, profitability, and market share trends. Further analysis includes frameworks to interpret a competitive environment using Porter's Five Forces and external trends using PESTLE analysis. Finally, we combine these approaches to understand a company's competitive strategy and its position relative to its industry peers.

LEARNING MODULE OVERVIEW

- Analysts use industry and competitive analysis to understand an industry's structural factors, relative competitive strengths and weaknesses, and their contribution to a company's economic profits.

- An analyst's first step in an industry and competitive analysis is to define the boundaries of the industry in question and its constituents. Industries are commonly defined as companies that sell similar products or services. However, this definition can be challenging to apply in practice, as companies can sell diverse products or services across many industries and product similarity is subjective.
- To define an industry, analysts can use third-party classification systems such as GICS but need to be aware of the methodologies and limitations of these systems.
- After defining the industry, the analyst surveys the industry by estimating its total annual sales, historical growth rate, and profitability metrics and then determining market shares and trends of key players.
- Industry metrics can be compared against broader economic trends during recessions or expansions to determine the industry's level of maturity, sensitivity to business cycles, and competitive rivalry.
- Analysts use Porter's Five Forces to understand an industry's structure, as defined by the level of competitive rivalry, through an assessment of the threat of new entrants, the threat of substitutes, customer bargaining power, and the bargaining power of suppliers.
- To supplement this research, analysts use PESTLE analysis to understand the external influences on and potential changes to an industry's growth rate and market share dynamics.
- To evaluate a company's competitive strategy, an analyst should determine whether the strategy creates a defense against industry forces and is aligned with the external forces acting on the industry, and whether the company has the resources and capabilities necessary to execute it.

LEARNING MODULE SELF-ASSESSMENT

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. When assigning a company to an industry, which of the following is a limitation of using third-party industry classification schemes?

   A. Currency differences

   B. Single-product companies

   C. Strictly hierarchical taxonomies

   Solution:

   C is correct. Commercial classification schemes such as GICS, IBC, and TRBC are examples of strictly hierarchical taxonomies that classify a company to a single group, regardless of whether the company sells multiple types of products or services.

2. A factor that determines sensitivity to the business cycle is *most likely*:

   A. customer migration to substitute products.

   B. interest rate exposure of the business model.

C. growth rates in line with broader economic activity.

Solution:

B is correct. Factors that determine sensitivity to the business cycle include the degree to which sales are discretionary or necessary for consumers, pricing power, the interest rate exposure of the business model, and whether the product is a durable or capital good versus a recurring purchase such as consumables and subscription services.
A and C are incorrect, as they reflect features of a mature industry but are not necessarily more cyclical.

3. A common measure of industry concentration is:

A. Porter's Five Forces.

B. the PESTLE framework.

C. the Herfindahl-Hirschman Index.

Solution:

C is correct. The Herfindahl-Hirschman Index (HHI) is a common measure of industry concentration that is calculated as the sum of the squares of competitor market shares. Porter's Five Forces model evaluates an industry's level of competitive rivalry and profitability. The PESTLE framework is more concerned with an industry's growth rate and market share dynamics.

4. Identify the following statement as true or false. Justify your answer.

Some of the forces included in Porter's Five Forces framework are the threat of new entrants, sensitivity to the business cycle, and the bargaining power of customers.

Solution:

False. Porter's Five Forces model uses the threat of substitutes, the threat of new entrants, the bargaining power of customers, and the bargaining power of suppliers to determine the rivalry among existing competitors. Sensitivity to the business cycle is *not* one of the forces.

5. The price competition historically demonstrated by automakers and aircraft manufacturers is *best* described as an example of:

A. the threat of substitutes.

B. the bargaining power of suppliers.

C. the rivalry among existing competitors.

Solution:

C is correct. Despite relatively low risks from the other Five Forces, automakers and aircraft manufacturers compete fiercely on price, offering promotions and generous financing and warranty terms in an attempt to capture every sale.

6. Standardization of a product will ____________ (increase/decrease/have no effect on) the bargaining power of customers.

Solution:

Standardization of a product will *increase* the bargaining power of customers. For example, oil refiners pay for crude oil based on its grade, not on its specific oil producer; crude oil within a grade is interchangeable.

7. Impression Ltd. is a fictional company that designs, manufactures, and sells skin care and beauty products. A PESTLE analysis for Impression Ltd. would *most likely* identify which of the following?

   A. A customer of Impression Ltd., a large e-commerce retailer, is acquiring a private-label manufacturer of skin care products.

   B. Increased discussion by legislatures around instituting an excise tax on disposable plastic packaging as part of a package of carbon taxes

   C. An online, direct sales competitor has launched a suite of photo filters that integrate with leading social media apps based on its beauty products.

   Solution:

   B is correct. PESTLE analysis is concerned with identifying and evaluating *external* forces on an industry, which include political forces such as a packaging tax. A and C are incorrect, as they are competitor moves within the beauty industry.

8. As a venture capital investor, you are on the board of Ridge Inc., a fictional company that is entering the auto industry, which is characterized by high capital intensity, minimal switching costs for customers because regulations require standardization of many features, and price consciousness of customers except among a relatively small percentage of affluent consumers.

   Based on these observations, recommend and justify a competitive strategy to the rest of Ridge Inc.'s board.

   Solution:

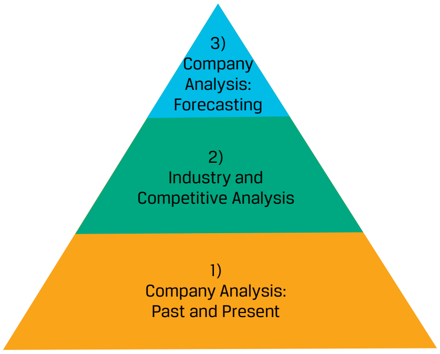
   A cost leadership strategy is one competitive strategy that might be successful for Ridge Inc. Since most customers are price conscious and face minimal switching costs between automakers or models, a low selling price can be an effective way to gain market share. Combined with high capital intensity, such a strategy might enable Ridge to operate at an economic profit if it produces and sells enough volume and maintains fixed-cost discipline to keep unit costs low. Other competitive strategies that might be successful for Ridge include a differentiation strategy aimed at luxury or performance customers.

2 USES OF INDUSTRY ANALYSIS

☐ describe the purposes of, and steps involved in, industry and competitive analysis

The next step in our company and industry analysis framework (Exhibit 1), introduced in the last module, is industry and competitive analysis, which involves the study of the drivers of an industry's size, profitability, and market shares and the evaluation of a company's competitive positioning in its industry.

Exhibit 1: Company and Industry Analysis Framework



3) Company Analysis: Forecasting
- Determine forecast objects and approaches
- Forecast revenue
- Forecast operating profitability and working capital
- Forecast capital investments and capital structure
- Evaluate key risks and uncertainties

2) Industry and Competitive Analysis
- Define Industry
- Analyze size, growth and character, profitability, market share trends
- Analyze industry structure and external influences
- Evaluate company's competitive strategy and positioning and determine if company has a competitive advantage

1) Company Analysis: Past and Present
- Determine business model
- Analyze revenue and revenue drivers
- Analyze operating profitability and working capital
- Analyze capital investments and capital structure (uses of capital and sources of financing)

Why Analyze an Industry?

Many industry participants have similar business models. They often compete in the same or similar product markets as sellers and in factor markets as buyers, so they tend to be exposed to the same demand and supply opportunities and risk factors. As will be discussed later with Porter's Five Forces model, these industry structural factors result in profitability differences by industry in the long run, with company-specific factors like business model variation, competitive strategy, size, and execution creating variance around the industry median.

McGahan and Porter (1997) estimated the relative importance of industry versus company-specific effects on both the emergence of economic profits (ROIC in excess of WACC) and their sustainability over time, finding that industry was the most important factor in the sustainability of economic profits. Importantly, they found that company-specific effects were much larger for low performers than for high performers in an industry, suggesting that industry and competitive forces can act as a ceiling on a company's returns but not as a floor.

A useful way to think about this dynamic is that competition "pulls" company profitability to an industry base rate over time. Industry and competitive analysis is used to estimate that base rate and its determinants, to form a forward-looking perspective on potential structural changes, and, finally, to analyze firms' relative strengths and weaknesses to determine their position against the base rate. It is important to emphasize, though, that these measures and outcomes are dynamic and that some companies have outperformed or underperformed their industry in economic profitability for many years.

Improve Forecasts

Competitive forces from industry incumbents, substitutes, suppliers, and customers discipline companies' prices and costs, market share, and thus profitability. By taking an industry perspective and better understanding these drivers—as well as building a database of past competitive actions and strategies by firms in an industry and their track records of success—analysts can sharpen their forecasts and alertness to what is most important. Guan, Wong, and Zhang (2014) found that sell-side investment analysts who covered both companies and their suppliers made more accurate earnings forecasts than those who did not cover the suppliers. Without a broader perspective, an analyst may underrate competitive forces and overrate the degree to which a company controls its destiny.

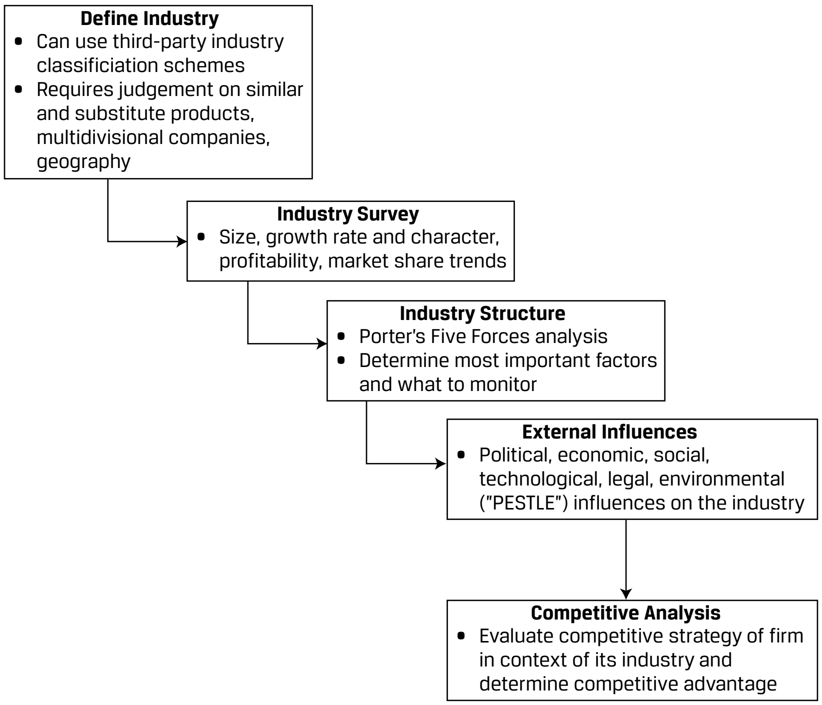
Identify Investment Opportunities

Industry analysis may uncover an attractive investment candidate that an analyst was previously unaware of or had not fully appreciated until assessing its strengths and weaknesses relative to industry peers and competitors. In addition, some investors may conclude that it is the industry, not any specific company, that they want exposure to, and in fact would like to diversify away the company-specific risks. These investors might take a basket approach by investing in several companies with position sizes scaled by size, liquidity, or relative attractiveness.

Industry and Competitive Analysis Steps

We approach industry and competitive analysis through the process shown in Exhibit 2, which begins with defining an industry and ends with evaluating a company's competitive positioning in the context of that industry.

Exhibit 2: Industry and Competitive Analysis Steps



QUESTION SET

1. Over the long run:

   A. business cycles result in profitability differences by industry.

B. company-specific factors such as competitive strategy result in profitability differences by industry.

C. structural factors result in profitability differences by industry.

Solution:

C is correct. Structural factors, such as Porter's Five Forces, drive profitability and profitability differences by industry in the long run. Company-specific factors result in company profitability levels around the industry median, while changes in the business cycle drive *short-run* profitability for both industries and companies.

2. Which of the following is an industry, *not* a company-specific, attribute?

A. Competitive strategy

B. Business model variation

C. Sensitivity to the business cycle

Solution:

C is correct. Sensitivity to the business cycle is an example of a factor that influences an entire industry.
A and B are examples of company-specific attributes, which also include company size and strategy execution.

3. Explain why industry has a significant effect on company growth and profitability.

Solution:

Companies in the same industry compete in similar product markets as sellers and in factor markets as buyers, so they tend to be exposed to the same demand and supply opportunities and risk factors. For example, while a restaurant company that sells chicken has a different product than a restaurant that sells pastries, both are competing for a similar customer need (prepared food), employ similarly skilled workers, and have capital equipment and real estate.

INDUSTRY CLASSIFICATION 3

☐ describe industry classification methods and compare methods by which companies can be grouped

An industry is commonly defined as companies that sell similar products or services, from the perspective of a customer. Challenges in defining an industry include whether to include substitutes, classifying companies that operate in multiple industries, geographical considerations, and making updates since business models are always evolving. To help analysts contend with these difficulties, third-party organizations maintain industry classification schemes that are widely used in investment management.

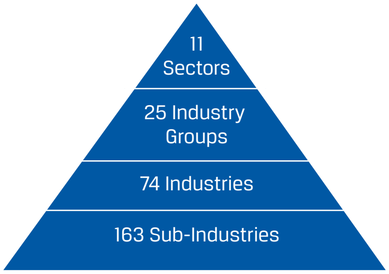
Third-Party Industry Classification Schemes

Early third-party industry classification schemes such as SIC, NACE, and ISIC were devised by government agencies, tended to be country-specific, and grouped companies by their production characteristics into industries such as agriculture, manufacturing, distribution, retail, and services. The schemes were not updated frequently and became less useful as new technologies and business models emerged and both issuers and investors became increasingly global. These legacy schemes used by investment analysts were largely replaced with commercial schemes such as the Global Industry Classification Standard (GICS) by MSCI and S&P Dow Jones Indices, Industry Classification Benchmark (ICB) by FTSE Russell, and The Refinitiv Business Classification (TRBC) by Refinitiv. These schemes are global, reviewed and updated at least annually, with new companies added more frequently, and group companies by the similarity of the products or services they sell—a "demand" approach rather than the "supply" approach of the legacy schemes. GICS and ICB cover public companies, while TRBC also covers private companies, non-profits, and government entities. Data aggregators (such as Bloomberg, FactSet, and CapIQ) and stock exchange operators (such as the London Stock Exchange) either use these industry classification schemes or have their own that are substantially similar.

GICS, ICB, and TRBC are strictly hierarchical taxonomies, analogous to the species, genus, family, etc., taxonomy used in biology. In each scheme, a company is classified to a *single* group in the lowest tier, with the groups themselves making up the higher tiers. In other words, continuing with the biology metaphor, a company is classified to a single species, which automatically assigns it to a genus, family, and so on. GICS, ICB, and TRBC have slightly different groupings, tiers, and names. Exhibit 3 shows the names and hierarchy of each scheme's tiers, the number of groups in each tier, and the composition of the highest tier.

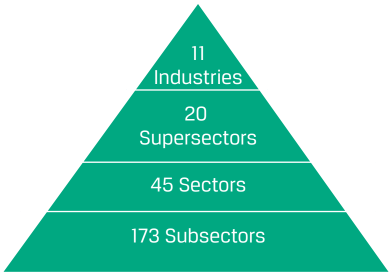
Exhibit 3: GICS, ICB, and TRBC Structures

GICS by S&P DJI and MSCI



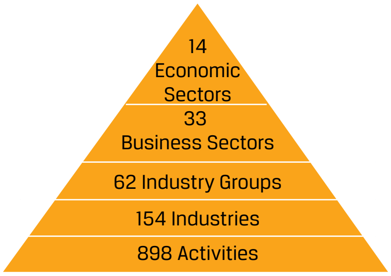
GICS Sectors
1. Energy
2. Financials
3. Materials
4. Information Technology
5. Industrials
6. Communication Services
7. Consumer Discretionary
8. Utilities
9. Consumer Staples
10. Real Estate
11. Health care

ICB by FTSE Russell



ICB Industries
1. Energy
2. Financials
3. Basic Materials
4. Information Technology
5. Industrials
6. Telecommunications
7. Consumer Discretionary
8. Utilities
9. Consumer Staples
10. Real Estate
11. Health care

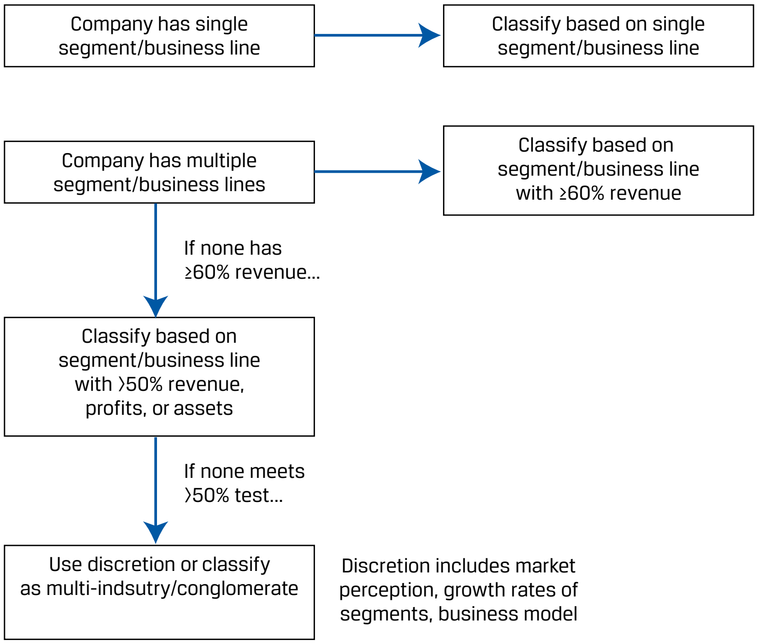
TRBC by Refinitiv



TRBC Economic Sectors
1. Energy
2. Financials
3. Basic Materials
4. Technology
5. Industrials
6. Communication Services
7. Consumer Cyclicals
8. Utilities
9. Consumer Non-Cyclicals
10. Real Estate
11. Healthcare
12. Institutions, Associations, and Organizations
13. Government Activity
14. Academic and Educational Services

GICS, ICB, and TRBC also have slightly different rules for companies operating in multiple industries, all of which use some discretion. While each scheme differs slightly, the general process is shown in Exhibit 4. The goal is to assign a company to a single grouping that describes the majority of its business.

Exhibit 4: Classifying Companies to Industries



Limitations of Third-Party Industry Classification Schemes

While these classification schemes are a useful starting place for research (and they have other uses like index construction and investment performance attribution, described elsewhere in the curriculum), they have four important limitations that analysts must contend with in doing industry research:

1. Groupings of companies with business model variations or that sell substitute products
2. The classification of multi-product companies
3. Geographical considerations
4. Changes in groupings over time that affect prior-period comparability of industry statistics

Analysts may find that industry groupings in the third-party schemes are either too narrow or too broad, as the scheme providers use discretion in constructing groups based on product similarity and business model. Therefore, analysts must either use classification tiers at a higher or lower level or modify the groupings based on their own discretion and purposes of analysis. On the one hand, many companies classified as "application software" sell products with completely different functionality and do not compete with one another—for example, Shopify, the Canadian company that sells front-end and payment processing software to small and mid-sized e-commerce companies, is grouped with Check Point Software, an Israeli security software company. On the other hand, as will be shown with Warehouse Club Inc., retailers are classified into numerous sub-industries and even across sectors, although many of their product offerings overlap and are thus competing for the same consumer dollar.

Because GICS, ICB, and TRBC are strict taxonomies, multi-product companies are assigned to a single grouping. This treatment can be a problem for an analyst identifying competitors if a competitor is a division of a company classified in a different industry or even sector. The most prominent examples are in technology. A large software market is cloud infrastructure and platform services (sometimes called “public cloud”), which are computing, storage, and other IT-related capabilities sold on a metered basis to clients over the internet. Large players include Amazon Web Services, Microsoft Azure, Google Cloud Platform, and Alibaba Cloud, which are all divisions of larger parent companies with other segments that differ materially by business model. While Amazon and Alibaba are classified in the same industry, Alphabet (parent of Google Cloud Platform) and Microsoft are each classified in different sectors. Amazon’s classification as a consumer discretionary retailer has long inspired consternation, as most of its operating profits come from its Amazon Web Services segment, not e-commerce.

With their global nature, GICS, ICB, and TRBC are useful for some sectors and industries that are in fact global (primarily those that manufacture physical goods and digital products), but they can be less useful for more service-oriented companies that compete nationally or even locally because of customer behavior or regulations that restrict competition. For example, only a small minority of people travel a significant distance from their home to receive healthcare services, and insurers (whether private or government) typically restrict the choice of provider geographically. Governments also regulate the practice of medicine through the licensing of facilities, physicians, and other practitioners. Therefore, the healthcare services industry is best examined on a national or regional basis; the economics of hospitals in the UAE have little relevance to those in France. However, this approach should not be taken for granted, as industries can globalize or localize based on business model changes. While media (pay TV) companies were once regional, the shift to internet-based streaming video has resulted in a global industry with new, larger players.

Finally, industry classification scheme providers change groupings and classifications, which can affect industry statistics. One of the largest changes, for example, was the creation of the Real Estate sector, composed primarily of Real Estate Investment Trusts carved out of the Financials sector. Additionally, public offerings and de-listings from acquisitions and bankruptcies change the composition of the groupings over time (introducing survivorship bias in historical returns statistics). Analysts should be aware of these issues, make sure that historical statistics are restated to conform to the latest groupings, and construct survivorship bias–free datasets if possible.

CASE STUDIES

Defining Warehouse Club Inc.’s Industry

Warehouse Club Inc. has a single line of business and would be assigned to the “Consumer Staples Merchandise Retail” sub-industry in GICS.

Sector	Industry Group	Industry	Sub-Industry	Other Sub-Industry Constituents
Consumer Staples	Consumer Staples Distribution & Retail	Consumer Staples Distribution & Retail	Consumer Staples Merchandise Retail	Costco, Walmart, BJ’s, Carrefour, Sun Art Retail Group, Coles Group, PriceSmart

While this classification is logical, the sub-industry includes companies outside Warehouse's domestic geography that are less relevant to its customers and does not include many companies in its domestic geography that sell similar products to the same types of customers. For example, grocers and drugstores are included in a different sub-industry (Food Retail and Drug Retail, respectively), and e-commerce retailers that it competes with are in a different sector entirely (Consumer Discretionary).

The challenge in placing Warehouse in an industry is that it sells several categories of consumer goods, both staples and discretionary goods like apparel, as do many other companies. Retailers also have varying business models—including membership-based warehouses, e-commerce, and co-locations with other services like pharmacies—and have varying location, assortment, and pricing strategies.

Elaine Nguyen chooses to define Warehouse's industry more broadly than its GICS sub-industry by taking a customer perspective: retailers of all kinds operating in Warehouse's domestic geography, except for automobile and auto parts retailers as Warehouse does not sell those products or even substitutes for them (and it is a large category of domestic retail sales). Nguyen also does not include restaurants and food services, instead considering them substitutes.

Alternative Methods of Grouping Companies

An industry or product approach to grouping companies is not the only grouping method and may be less useful in contexts outside industry analysis. Other approaches that are used in contexts such as index construction and investment performance evaluation include the following:

- Geography, in which companies are classified by country and then countries are aggregated into categories such as developed, emerging, and frontier markets. Classification by country is typically by the country where the issuer is incorporated, the country of the primary listing of its equity securities, the location of its headquarters, or market perception. Note that classification by the geographic composition of revenue is generally *not* the approach taken, though this aspect may be the foremost concern for an analyst. For example, Toyota is universally classified as a Japanese company. However, its largest market by revenues is North America, which is also the location of most of its assets. The classification of developed, emerging, and frontier markets is more controversial, and third parties like index providers use greater discretion. It is not a quantitative determination but considers variables such as the size and liquidity of equity securities markets in the country, income per capita, and legal restrictions on foreign investment.
- Sensitivity to the business cycle, with groupings such as "defensive" and "cyclical." Defensive companies are those whose sales growth, profitability, and valuations are less affected by changes in broad macroeconomic activity (e.g., GDP growth), while the opposite is true of cyclicals. Generally, this classification is done by grouping entire sectors from the industry classification schemes discussed earlier. Sectors such as consumer staples, healthcare, and utilities are considered defensive, while financials, basic materials, consumer discretionary, and industrials are considered cyclical. This grouping method can be combined with other methods—such as a geographic approach (e.g., Japanese cyclicals), because countries may vary with respect to their current position in the business cycle, or credit ratings (e.g., Japanese cyclicals with high-yield credit)—to further isolate exposure to certain risk factors.

- Statistical similarities, or the use of clustering analysis to group companies based on similarities of financial ratios and market data or co-movements of their securities' investment returns. This approach includes grouping by size according to market capitalization or by other characteristics such as the following:
  - Valuation ratios
  - Growth rates of sales or earnings
  - Profitability ratios
  - Statistics based on price performance such as volatility and momentum
- ESG characteristics, such as the ratio of carbon emissions to revenues, measures of board and executive personnel diversity, and exposure to certain businesses such as tobacco and gambling. These metrics can be aggregated into composite ESG ratings or scores that enable cross-issuer comparability.

These groupings are usually relative (e.g., companies with dividend yields at or above the 75th percentile as of the measurement date) and tend to show far more turnover in their constituents than groupings based on industries and countries, because these statistics are less stable by company and companies' rankings change.

QUESTION SET

1. Contrast the Global Industry Classification Standard (GICS) commercial scheme with the Standard Industrial Classification (SIC) third-party classification system in terms of geographic coverage, update frequency, and the addition of new companies.

   Solution:

   The GICS commercial scheme is determined on a global basis, updated at least annually, and adds new companies frequently, while the SIC is US-only, infrequently updated, and does not regularly add new companies.

2. The GICS, ICB, and TRBC commercial industry classification schemes suffer from which common problem when used by analysts?

   A. They quickly become out of date.

   B. They cover a limited number of countries.

   C. They often classify multi-industry companies inaccurately.

   Solution:

   C is correct. Commercial industry classification schemes are strictly hierarchical and assign each company to a single industry. Multi-industry companies are classified using evaluations of segment revenues, profits, or assets or the discretion of the scheme creator. This approach often results in classifications that are different from analyst expectations or in "multi-industry" classifications that are challenging to use in practice.

3. Identify whether each of the following sectors is generally considered "cyclical" or "defensive."

I. Consumer Staples	Cyclical	Defensive
II. Consumer Discretionary	Cyclical	Defensive
III. Energy	Cyclical	Defensive

IV. Healthcare	Cyclical	Defensive
V. Utilities	Cyclical	Defensive
VI. Financials	Cyclical	Defensive
VII. Industrials	Cyclical	Defensive
VIII. Technology	Cyclical	Defensive
IX. Materials	Cyclical	Defensive
X. Real Estate	Cyclical	Defensive

Solution:

I. Defensive
II. Cyclical
III. Cyclical
IV. Defensive
V. Defensive
VI. Cyclical
VII. Cyclical
VIII. Defensive
IX. Cyclical
X. Defensive

4 INDUSTRY SURVEY

☐ determine an industry's size, growth characteristics, profitability, and market share trends

After defining an industry, the next step in industry and competitive analysis is to survey the industry by estimating its size, calculating its historical growth rate, evaluating the character of that growth rate, measuring its profitability, and identifying major industry players and market share trends over time. This industry survey provides a basis for the industry evaluation in the next step and makes an analyst aware of the main issues and opportunities.

Industry Size and Historical Growth Rate

Industry size is typically measured by total annual sales from the product or customer perspective, which is not necessarily all sales of each industry constituent. For example, Amazon's retail segment sales would be included in sizing the retail industry, while sales for its Amazon Web Services segment would not. The growth rate is calculated either as year-over-year rates each year or as a compounded annual growth rate over a multi-year period. If possible, industry growth should be broken out by contributions from volume and price/mix drivers.

Except for some industries that are dominated by large, publicly traded companies (e.g., autos, smartphones, airlines, pharmaceuticals), industry size will often include a potentially large amount in sales from private companies, sometimes small businesses, for which data are unavailable or impractical to tabulate. It's common practice to estimate industry size for these industries using economic indicators published by government agencies, data from third-party consultancies that use surveys, or industry data from issuers' investor presentations based on proprietary sources. These all need to be corroborated for reasonableness by the analyst. For Warehouse Club Inc., for example, we use retail sales published by a government agency for industry size, because the retail industry includes hundreds of thousands of small retailers.

Characterizing Industry Growth

The pattern of historical growth of an industry can be used to characterize it based on the magnitude of its growth rate and sensitivity to the business cycle. One approach to characterizing an industry's growth rate is a style box, as shown in Exhibit 5. Essential to this approach is comparing the industry's historical growth rate with a broader measure of economic growth during recessions and expansions.

Exhibit 5: Industry Growth Style Box

	Mature	Growth
Defensive	–Utilities –Beverages –Pharmaceuticals	–Biotechnology –Software –Gaming
Cyclical	–Crude oil –Natural gas –Freight transportation	–Semiconductors –Fintech –Digital advertising

Source: Authors' analysis.

Growth industries include those that have not yet reached full saturation or penetration of their total addressable market, and thus have idiosyncratic growth drivers separate from broader economic growth. Growth industries are often based on or benefit from an emerging technology. Key questions for analysts related to growth industries include whether historically high growth rates will persist and what the peak penetration rate is—which are more challenging to answer the "newer" the industry is. Conversely, mature industries have fully penetrated their addressable market and have growth rates either in line with broader economic activity (approximately zero) or declining as customer demand migrates to a substitute. Key issues for investors in a mature industry include monitoring for disruptive threats, changes in competitive intensity, and the speed of decline.

Sensitivity to the business cycle is driven by the business models of companies in an industry and correlates with industry maturity. Factors that determine sensitivity include the degree to which sales are discretionary or necessary for consumers, prices, the interest rate exposure of the business model, and whether the product is a durable or capital good versus a recurring purchase such as consumables and subscription services. Although most investors expect greater variability in returns from firms in cyclical industries, differing investor views as to the length and magnitude of such cycles lead to wide variations in the valuations of issuers over time.

Although commonly used by market participants, these distinctions can sometimes be of limited usefulness. For example, a severe economic downturn is likely to have a negative impact on all firms, so differences will be a matter of degree but not categorical. Second, companies in different stages of the life cycle within an industry can differ materially in terms of growth, defensiveness, and cyclicality; for example, growth firms based on an emerging technology may exist within a cyclical industry that is far less affected by an economic downturn.

Industry Profitability Measures

The best measure of industry profitability is a time series of the distribution of returns on invested capital (e.g., 25th, 50th, 75th percentiles), which captures after-tax operating profits for each dollar of invested capital and is agnostic about capital structure. However, unless the industry is dominated by public companies, obtaining these data is seldom feasible.

Two common methods to overcome this challenge in practice are to measure the profitability of the publicly traded players and assume that the profitability of private competitors is similar—either using various sources to estimate the profitability of non-public companies (e.g., for state-owned energy companies, there are estimates of their production costs per barrel, production, and capital expenditures, and prices are generally uniform throughout the industry) or obtaining data from a government agency or third-party consultancy. For Warehouse Club Inc., the same government agency that publishes retail sales data each month also does an annual survey of retailers' gross margins and operating expenses, which can be used to calculate industry operating margins.

Clearly, a profitable industry is preferable to an unprofitable one, but often more important than a point estimate of profitability is the time series: is industry profitability rising or falling? If there is a discernible trend, it will be an important point of investigation for the analyst.

Market Share Trends and Major Players

Finally, market shares are measured by expressing industry participants' annual revenues as percentages of the industry size each year. Given the aforementioned measurement problems with estimating industry size, market shares should be interpreted as a range rather than a definitive point estimate. Again, more important than each company's market share is the trend over time, which can reveal whether customers have judged the company's products superior to competitors'. An important consideration is acquisitions, particularly large ones; a company buying a competitor is a market share gain, but more important is whether the company is increasing or decreasing its share on an organic (non-acquired) basis.

An additional consideration in this analysis is the degree of industry concentration. Lower concentration—that is, many small competitors in the market—is usually associated with a high degree of competitive intensity unless the industry is very service-oriented, is local in nature, or has high product differentiation. Rising concentration (consolidation) is associated with falling competitive intensity and higher profitability. A common measure of industry concentration is the **Herfindahl-Hirschman Index (HHI)**, calculated as the sum of the squares of competitor market shares:

$$HHI = \sum_{i=1}^{\infty} s_i^2 ,$$

where s is the market share of market participant i stated as a whole number (i.e., 50% share = 50, not 0.50).

A market consisting of four firms with shares of 30, 30, 20, and 20 has an HHI of $30^2 + 30^2 + 20^2 + 20^2 = 2{,}600$. The maximum HHI is for a monopoly, with a value of $100^2 = 10{,}000$. Antitrust regulators in some countries consider markets with an HHI between 1,500 and 2,500 moderately concentrated and consider markets with an HHI over 2,500 highly concentrated. A rule of thumb is that acquisitions in highly concentrated markets that increase the HHI by more than 200 points are often subject to regulatory challenges.

CASE STUDIES

Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd.

Industry Size and Growth Rate

Elaine Nguyen sizes the US retail (excluding autos) industry using three measures: total annual revenues, annual revenues on a per capita basis, and the number of retailers. For all these measures, Nguyen uses data from the US Census Bureau, since the industry consists mostly of small businesses with no public filings and the data are high quality; the Bureau has been collecting such data since 1952.

US Annual Retail Sales ex Autos (Bars, left) and Growth Rate (Line, right)

At the end of 2X19, there were 556,531 firms in the industry, but over 92% of them employed fewer than 20 people. Just 470 large firms with over 5,000 employees accounted for 62% of total industry employment. Of these large firms, 144 were publicly traded.

US Retailers (ex autos) as of 2X19	
< 20 employees	516,766
Between 20 and 5,000 employees	39,295
> 5,000 employees	470

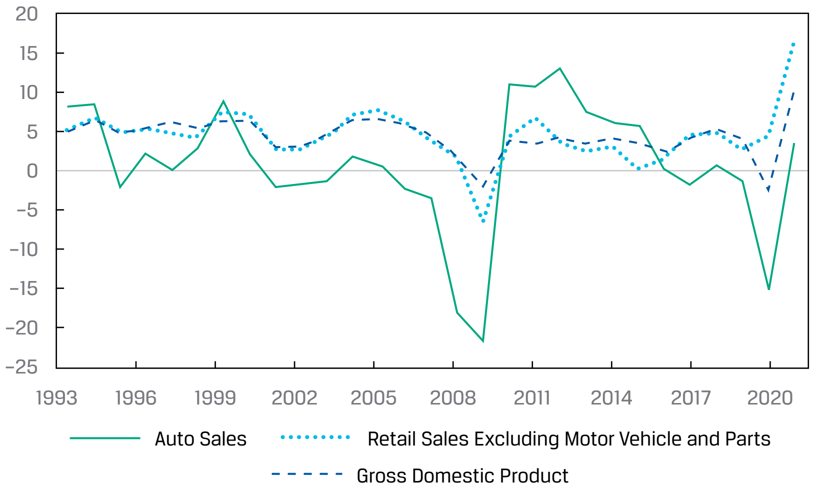
US Retailers (ex autos) as of 2X19	
Total	556,531
Note: public companies	144

Character of Industry Growth

The retail industry has existed for hundreds of years in the USA, though there have been many changes in both form (e.g., department stores, discounters, e-commerce) and the products sold. Comparing industry sales growth with GDP growth since the early 1990s shows that the two are tightly related. However, the industry does not show extreme cyclicality, unlike the auto industry.

Elaine Nguyen classifies the industry as mature and moderately cyclical.

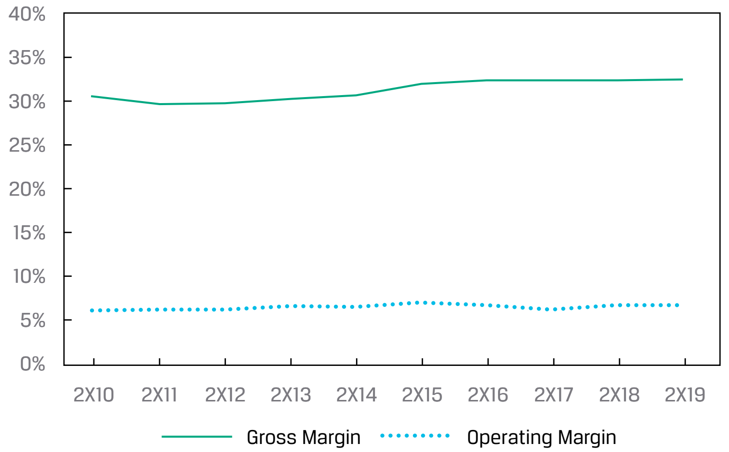
Annual Growth Rates: US Retail Sales ex Autos, Autos, and GDP



Industry Profitability

Nguyen analyzes industry profitability in two ways: (1) the gross and operating margins of all retailers (except autos) in the USA using Census Bureau data and (2) the profitability of the 10 largest publicly traded retailers in the USA by 2X19 sales. Because of segment disclosure changes, M&A, and other corporate transactions, Nguyen examines the last 5 rather than 10 years of data.

Gross and Operating Margins of US Retailers ex Autos

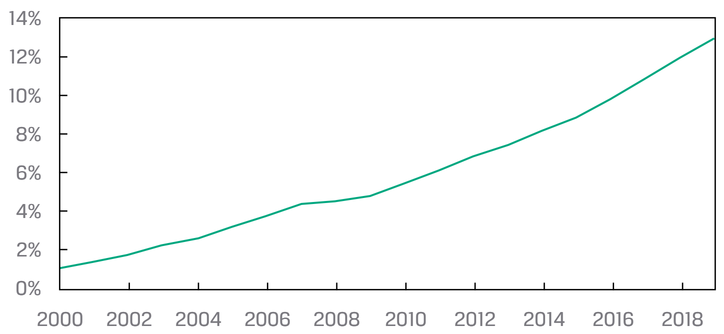


Operating Margins	2X15	2X16	2X17	2X18	2X19
Walmart (US segment)	6.4%	5.8%	5.3%	5.2%	5.1%
Walmart (Sam's Club segment)	3.2%	2.9%	1.5%	2.6%	2.8%
Amazon.com (North America segment)	2.2%	3.0%	2.7%	5.1%	4.1%
Costco Wholesale (US segment)	2.7%	2.7%	2.8%	2.7%	2.7%
The Home Depot, Inc.	13.3%	14.2%	14.5%	14.4%	14.4%
Kroger Co.	3.3%	3.0%	2.1%	2.1%	1.8%
Target Corp.	6.9%	7.1%	5.8%	5.5%	6.0%
Lowe's Companies, Inc.	8.4%	9.0%	9.6%	7.9%	8.8%
Albertsons Companies, Inc.	0.7%	1.1%	0.3%	1.0%	1.5%
Ahold Delhaize (US segment)	4.0%	3.9%	4.1%	4.5%	4.3%
The TJX Companies, Inc. (US segments)	13.0%	12.8%	11.8%	11.8%	11.6%

Market Share Trends

The most significant market share trend in US retail is the shift toward e-commerce, which has been ongoing since the late 1990s. This shift has accrued largely to Amazon.com, the pioneer in US e-commerce, with its share of e-commerce at roughly 40% (Amazon has a large third-party merchant business but does not disclose GMV, so its exact share is unknown).

US E-Commerce % of Retail Sales ex Autos and Petrol



The industry is fragmented, with the top 10 largest firms holding less than 30% market share, the 10th largest holding less than 1%, with a large amount of share held by over 500,000 small retail businesses. However, the top 10 largest firms have gained share (~2%, from 26.3% in 2X15 to 28.6% in 2X19).

US Net Sales	2X15	2X16	2X17	2X18	2X19
Walmart US and Sam's Club	355,206	365,198	377,693	389,505	399,796
Amazon.com*	63,708	79,785	106,110	141,366	170,773
Costco Wholesale (US segment)	84,351	86,579	93,889	102,286	111,751
The Home Depot, Inc.	80,550	86,615	92,413	99,386	101,333
Kroger Co.	109,830	115,337	123,280	121,852	122,286
Target Corp.	73,717	69,414	71,786	74,433	77,130
Lowe's Companies, Inc.	59,074	61,311	63,263	65,872	67,147
Albertsons Companies, Inc.	58,734	59,678	56,925	60,535	62,455
Ahold Delhaize (US segment)	42,482	42,946	43,194	44,174	44,841
The TJX Companies, Inc. (US segments)	23,899	25,651	27,365	29,845	32,021

US Net Sales	2X15	2X16	2X17	2X18	2X19
Total top 10	951,551	992,514	1,055,918	1,129,254	1,189,533
Share of US retail sales ex autos	26.3%	26.9%	27.4%	27.9%	28.6%

**Amazon.com's "North America" segment only.*

QUESTION SET

1. Identify the following statement as true or false. Justify your answer.

   Industry size is typically measured by the total annual sales of all industry constituents.

   Solution:

   False. Industry size is typically measured by the total annual sales from the product or customer perspective, which is not necessarily the sum of total sales of each industry constituent as some constituents may have segments in other industries.

2. A section of the fast-food market consists of 10 firms operating 1,840 restaurant locations across North America and currently has a Herfindahl-Hirschman Index (HHI) of 1,516. The market leader announces plans to acquire its closest competitor and commits to not closing any locations at either company. As a result, the HHI will *most likely*:

   A. decrease.

   B. not change.

   C. increase.

   Solution:

   C is correct. The Herfindahl-Hirschman Index (HHI) is calculated as the sum of the squares of competitor market shares. In this example, the market share of the leading company would increase and the number of competitors would decrease, which would increase the HHI. The number of store locations does not affect the HHI.

3. Identify the following statement as true or false. Justify your answer.

   Companies in different stages of the life cycle within the same industry can differ materially in terms of cyclicality.

   Solution:

   True. An industry is defined as companies with similar products or services from the perspective of a customer. A new entrant in an industry may have low but increasing market share, which can serve as a growth driver in addition to an economic expansion. The new entrant may therefore be less affected than a mature company by a recession.

Please refer to the Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd. earlier in the lesson to answer the following questions.

4. Amazon.com's change in US retail market share from 2X15 to 2X19 is *best* described as an example of:

   A. a disruptive threat.

   B. achieving peak penetration rate.

   C. saturation of its addressable market.

   Solution:

   A is correct. As the pioneer in US e-commerce, Amazon achieved roughly 40% market share by using Amazon.com as an online shopping platform. Therefore, Amazon.com is an example of a disruptive threat to the established retail industry. Based on US net sales from 2X15 to 2X19, it is uncertain whether Amazon achieved its peak penetration rate or saturated its addressable market, as it continued to increase its market share component of the top 10 retailers' share from 7% to 14%.

5. Current economic forecasts call for an increase of 3% in US GDP, an inflation rate of 1.5%, and decreasing consumer sentiment. Therefore, over the next fiscal year, Nguyen would *most likely* expect Warehouse Club Inc.'s total sales to:

   A. decrease.

   B. increase 2%–5%.

   C. increase 5%–8%.

   Solution:

   B is correct. Nguyen classifies the industry as mature and moderately cyclical, and Warehouse Club Inc. operates a single line of business with no new products or services defined. As a result, Warehouse Club Inc.'s total sales would move in line with broader US GDP growth as a stable company in a mature industry.

5 INDUSTRY STRUCTURE AND EXTERNAL INFLUENCES

☐ analyze an industry's structure and external influences using Porter's Five Forces and PESTLE frameworks

Recall from earlier lessons in economics that Porter's Five Forces is a framework for assessing industry structure that determines an industry's long-run profitability measure by its returns on invested capital.

If some or all of the five industry forces are intense, almost no company earns attractive returns on capital, while benign forces lead to highly profitable industries. From the industry survey work in the previous lesson, we may have already measured the industry's profitability on a historical basis. The analysis of structural forces is aimed at not simply measuring profitability but rather characterizing its drivers in a qualitative fashion and improving alertness to key issues to monitor that can affect industry profitability. For new industries for which data are unavailable or which are still in an early, unprofitable stage, the analysis of industry structure can help point to potential profitability levels.

Assessing the Five Forces: A Checklist Approach

The assessment of the five forces is similar to the determination of a company's business model, from the prior module, in that it is a qualitative research investigation that is difficult to generalize. We provide illustrative questions to answer as an analyst and offer examples of real industry observations for each question.

Assessing the Threat of New Entrants

- Have there been significant numbers of new industry entrants in recent history?
  - In the USA, more than 50,000 new restaurants are opened per year. Conversely, from 2010 to 2021, fewer than three new commercial bank charters were issued per year in the USA.
- Are there network effects with the industry's product or service, such that its value to users is greater as the number of users increases and a new entrant would have a low-value product that has to achieve scale?
  - Payment networks such as Visa, Mastercard, JCB, and China UnionPay enable payments between hundreds of millions of cardholders and millions of merchants by connecting thousands of financial institutions on their network.
- Do incumbents benefit from economies of scale from significant fixed costs in the cost structure, such that a new entrant would require time and unprofitable economics to reach competitiveness?

- Cloud computing services require enormous fixed assets in terms of servers and networking equipment. These services have virtually no variable cost, however, and large incumbents have decreased unit prices substantially over time while gaining scale.

- Do incumbents benefit from economies of scope across multiple business lines—from increased convenience in terms of customers and market share and/or from joint costs that can be leveraged across the businesses?

  - Warehouse Club Inc. operates pharmacies and gas stations, which leverages real estate costs and drives increased foot traffic. Many commercial banks operate consumer banks, asset and investment management, broker-dealers, and investment bank businesses, as there are common customers and joint operating costs.

- Are customers loyal to brands?

  - A common threat to branded consumer merchandise sold to retailers is merchandise manufactured by contract manufacturers at the direction of the retailer and sold under retailer-owned "private label" or "generic" brands. In some categories, such as dairy milk and packaged baked goods, private label has over 50% volume share, which has risen over time. Consumers do not appear to be driven by brands in these categories. However, energy drinks, alcoholic beverages, and tobacco products have a very low private label market share over time, with consumers making repeat purchases of the same brand. Note that an important consideration for the value of brands is the legal environment that protects trademarks, trade dress, copyright, and other intangible assets in a jurisdiction. Without these legal protections, brands and their likeness can be imitated by competitors.

- Are there significant switching costs for customers in terms of time, training, or replacement?

  - So-called infrastructure software such as SAP for businesses requires lengthy implementation processes led by third-party consulting firms, and connections to other mission-critical software systems need to be established. Switching from SAP to another provider is a costly and lengthy endeavor. Conversely, customers can and do switch between food delivery apps, as minimal useful information is needed (or stored) to switch apps (payment information).

- Do incumbents have exclusive or preferential access to hard-to-obtain inputs or end-customer interfaces?

  - In the beverages business, incumbents have shelf space arrangements with retailers, often own distribution assets, and in some countries have the same arrangements with restaurants and pubs. This advantage is a potent deterrent for the beverage companies' customers to start carrying new entrants' products.

- Are there government policies that restrict or slow industry entry?

  - These policies can take many forms, such as regulations that only larger companies can afford to comply with, subsidies for domestic companies, patents, and licenses.

Note that experience and expertise among incumbents are not sustainable barriers to industry entry. Entrants can imitate, poach employees, purchase inputs and services from the incumbents' suppliers, and engage with industry consultants to obtain industry expertise.

Assessing the Threat of Substitutes

- Is there another product or service that fulfills the same or similar customer need? Is it lower priced?
  - Video conference calls and collaboration software are a low-price substitute for business travel and office real estate.
- Can customers go without the industry's product?
  - Aesthetic and elective medical procedures can be canceled or deferred without loss of function for a patient.
- Is the substitute product improving in performance relative to the industry's product?
  - While digital advertising was rudimentary in the early 2000s, with pop-ups and display ads, and was not a material threat to television advertising, digital advertising improved to include highly targeted videos and advertising disguised as content from third-party influencers and has taken market share from television advertising.
- Is it difficult for customers to switch to a substitute?
  - In most countries, prescriptions for a branded pharmaceutical are automatically substituted with a generic at a pharmacy if there is one approved.

Assessing the Bargaining Power of Customers

- Are there few, concentrated customers for the industry?
  - For many electronic component companies, mobile device makers such as Apple, Samsung, and Xiaomi account for a significant percentage of their revenues.
- Are the industry's products standardized or undifferentiated?
  - Crude oil is graded (e.g., Brent, West Texas Intermediate) by its density and sulfur content, which determine its cost and the effort to refine it into products such as petroleum. Oil refiners pay for crude oil based on its grade, not by specific oil producer; crude oil within a grade is interchangeable.
- Are the industry's products critical for customers?
  - Pesticides and seeds, especially those made with biotechnology, are sold at high prices but are vital to sustaining high agricultural yields and the high prices are outweighed by the cost of crops lost to disease and insects.
- Are the industry's products a large part of customers' budget?
  - Customers may be more motivated to seek alternatives and negotiate for products that account for a large part of their budget and for which quality is not a paramount concern. For livestock farmers, the cost of feed

accounts for over 60% of annual operating costs, while pharmaceuticals and vaccines for livestock account for a small percentage but come with a high cost of failure, which has given animal-pharmaceutical companies a strong bargaining position.

- Can customers backward-integrate—that is, "build" rather than "buy" what the industry is selling?
    - Streaming video services first licensed content from existing studios and media companies, but over time they vertically integrated by producing their own video content.

Assessing the Bargaining Power of Suppliers

- Are there few, concentrated suppliers for the industry?
    - Apple's "A" and "M" series processors for its iPhones and MacBooks are manufactured exclusively by TSMC. While Apple has more supplier choice for its other components, it is beholden to a single company for this crucial component.
- Are there significant costs for switching suppliers?
    - Asset managers need performance data and analytics on equity, fixed income, and other indexes for investment processes and client reporting of benchmarks. Indexes are owned by a small number of companies and charge licensing fees for data permissions and commercial use. An, Benetton, Song, and Index Providers (2021) found that managers of indexed exchange-traded funds paid index licensing fees equal to 35% of their management fee revenue. While asset managers can switch because each provider offers similar catalogues of indexes, doing so would require changing investment processes, client reporting, and conversations with clients—and can look bad to prospects.
- Are suppliers differentiated or specialized?
    - Healthcare providers, airlines, investment banks, technology companies, and professional services firms all require skilled, credentialed, or licensed labor, for which there is limited supply.
- Are there substitutes for the products from suppliers?
    - Retailers can substitute branded merchandise for private label merchandise from contract manufacturers.

Assessing Rivalry among Existing Competitors

- Is there a history of price competition among competitors?
    - Despite a low risk from every other industry force, automakers and aircraft manufacturers compete fiercely on price, offering promotions and generous financing and warranty terms to try to capture every sale.
- Are there numerous, roughly equal-sized competitors?
    - There are approximately a dozen automakers with an annual global market share over 1%. However, the largest company (Toyota) has had less than 15% market share for many years.
- Are products differentiated?

- Pharmaceuticals without a patent (generics) are legally required to be chemically equivalent to their originator drug in order to be approved by regulators in an abbreviated process that doesn't involve a clinical trial. Pharmacies are indifferent as to whether an approved generic is sold by one company or another and manufacturers' costs are largely fixed, so the companies compete fiercely on price for volume. Price competition increases with the number of approved generics per originator drug; Dave, Hartzema, and Kesselheim (2017) found that a drug with one generic competitor had prices that were, on average, six times higher than the prices of one for which there were nine generic manufacturers.

- Are there high barriers to exiting?

  - If exiting an industry is difficult or costly, such as shutting down natural resource–producing assets, companies with low profitability may remain in the industry for much longer than in other industries, which can result in lingering competition.

- Is industry growth slow?

  - If penetration of the addressable market is low, companies may not feel a need to compete for customers using price and other economic terms. This was the case for mobile wireless plans until the mid-2010s, when penetration plateaued and companies began engaging in aggressive promotional activity in competition for one another's customers.

After completing the evaluation of industry structure, an analyst must judge which are the most and least important factors. It is often clear that one or two forces are strongest, but there isn't a quantitative test or threshold by which to judge definitively. Posing scenarios to experienced industry insiders is often useful, such as asking, "What do you think would happen if company X increased its prices by 10% tomorrow?" If their answer is that a large percentage of customers are willing and able to switch to a competitor or substitute—perhaps it has happened before—it's likely that customer bargaining power and the threat of substitutes warrant further investigation.

CASE STUDIES

Porter's Five Forces Analysis for US Retail (ex autos)

Threat of New Entrants: Very High

Opening a retailer, especially an e-commerce retailer that utilizes third-party merchant services, is relatively easy and common. In the USA, retailers are the most common type of business formed—by a wide margin, with over 40,000 new firms filing formation papers each month (the next most common is transportation and warehousing, with around half the number of retailer formations). Customers can easily switch retailers, as most do not have an ongoing relationship such as a subscription fee or contract, and there are minimal regulatory barriers such as licenses and patents.

Threat of Substitutes: Low

Broadly speaking, the substitute for retail is consumer services (e.g., restaurants, travel, healthcare), which also includes digital services (such as streaming video subscriptions and gaming). Most categories of goods, however, are not easily replaceable with services (e.g., apparel and home décor), or they enjoy a cost advantage because of lower labor intensity (e.g., fresh food). There have been periods when services grew faster than goods, but as shown in an earlier exhibit, retail sales have grown essentially in line with US nominal output/income. Retail is arguably one of the oldest industries, and while it has evolved over time, it has yet to be replaced.

Bargaining Power of Customers: Moderate

Retail customers are highly fragmented, with each consumer representing a distinct decision-maker, as there are generally no group purchasing organizations. However, many products are sold by many retailers, and the internet has enabled easy comparison shopping at retailers. Customers are price sensitive with respect to identical products sold by different retailers.

Bargaining Power of Suppliers: Low to Moderate

Key suppliers for retailers include manufacturers of goods, employees, and lessors of retail or fulfillment space. In most cases, numerous options are available. However, branded goods are sold exclusively by a sole manufacturer that may impose high prices and other economic terms, like shelf space and visibility, and may want to sell only to certain retailers (e.g., makers of luxury goods may not sell to discount retailers in order to maintain exclusivity).

Rivalry among Existing Competitors: High

Given the sheer number of similarly sized firms selling similar or identical products, retailers compete fiercely, often with price promotions and discounts as price is one of the few ways they can lure customers away from competitors.

External Influences on Industry Growth

Analyzing the structure of an industry primarily involves looking *inside* the industry and at close adjacencies such as customers, suppliers, and substitutes. It is important for analysts to look *outside* the industry for factors that influence the industry's economic outcomes. One framework for this purpose is a **PESTLE analysis** of the *p*olitical, *e*conomic, *s*ocial, *t*echnological, *l*egal, and *e*nvironmental influences on an industry. While Porter's Five Forces is concerned with the determinants of industry profitability, a PESTLE analysis is more concerned with an industry's growth rate and market share dynamics. For example, athletic apparel companies such as lululemon athletica inc. have long benefited from increasing consumer interest in fitness and wellness, a social influence on the apparel industry. This trend is not apparent from Porter's Five Forces but is obviously important for growth in athletic apparel. When the trend eventually plateaus or declines, it's likely to result in declining sales growth or sales for lululemon, so analysts carefully monitor the trend using a variety of metrics. PESTLE analysis is a framework for identifying "themes" or "narratives" that investors may take a perspective on and desire exposure to.

While analysts must look broadly for influences, several factors make this search easier in practice. First, external influences often do not change very quickly (though their magnitude and direction can), so this analysis does not need to be done every quarter. Second, analysts develop industry expertise, and those in an institutional setting can access industry specialists who offer their perspectives and surveillance of industry trends as a paid service. Finally, it is usually the case that some influences are far more important than others for a given industry.

Political Influences

Political influences on an industry include changing fiscal and monetary policies, governments' direct selling and purchasing activities, regulatory changes, and geopolitical conditions and actions. We highlight three sectors and their constituent industries for which political influences are often significant: energy, healthcare, and defense.

The energy sector is subject to three important political influences: the political appeal of low and stable energy prices in the short run, climate agreements and legislation calling for emissions reductions, and the actions of OPEC (Organization of the Petroleum Exporting Countries). Most governments desire low energy prices in the short run to maintain their appeal to consumers and businesses that have inelastic demand with respect to price. However, in the long run, higher energy prices (perhaps because of a carbon tax) might be preferable in order to curb demand and meet emissions targets, for which the consumption of fossil fuels is the main driver. Investors and producers may worry about the longer-term outlook, since governments are intending to shrink fossil fuels as a percentage of the energy mix, but reductions in fossil fuel production are at odds with low energy prices in the short run. Finally, energy companies owned by member states of OPEC produced 35% of crude oil globally in 2021 (*BP Statistical Review of World Energy*) and set production quotas as a bloc. Actions taken by OPEC significantly affect global and regional energy prices. OPEC members may choose to keep prices low enough to slow the transition to renewables, exploiting governments' need for low energy prices in the short run.

Governments are the largest buyers of healthcare products and services and, in most cases, are the exclusive buyers of defense goods and services globally. For political appeal and to shore up fiscal deficits, governments have at times pursued reforms such as expansions of healthcare coverage and subsidies, price controls or reductions, and restrictions on certain healthcare services or products.

Defense spending is determined by countries' geopolitical objectives, perceived threat environment, and alliance commitments. From the late 1990s to 2010, global military spending rose markedly, increasing as a share of global GDP, which benefited defense companies. Future defense spending depends on geopolitical events and other competing budget priorities. Finally, governments also extensively regulate healthcare and defense, the stringency of which changes over time. From 2000-2007, several high-profile pharmaceutical scandals led to a period of heightened regulatory scrutiny and a declining number of drug approvals. This scrutiny was relaxed by the mid-2010s. Countries are selective about which countries they permit domestic defense companies to do business with, and the contracting process is highly complex, involving third-party costs and performance audits.

Economic Influences

Economic influences on an industry include changes in GDP or personal income, inflation, interest rates, and exchange rates. These influences can be related to the business cycle (cyclical) or structural, such as how longer-run population and productivity growth rates have differed across countries, which affects the sales and costs of companies operating in different countries. Economic influences are most important for cyclical sectors such as financials, basic materials, and consumer discretionary, and exchange rate changes are most important for industries and companies operating in multiple currencies—especially if there are large currency compositional mismatches in revenues and costs and if the company does business in an illiquid currency with more volatile exchange rates.

The auto industry is among the most affected by economic influences, because it sells high-priced durable goods, the purchase of which can be delayed (consumer keeps current car longer) or substituted (used car) based on consumer confidence. Structural economic influences are important for many multinational companies that have sought to increase exposure to faster-growing emerging markets over time. As of 2022, over 40% of Nestlé SA's sales were in emerging markets, which have long shown faster volume growth than its developed-market sales.

Social Influences

Social influences include cultural and consumer trends, demographic changes such as the relative growth rates among different population cohorts, and changes in lifestyles. These influences are not entirely external to an industry, as companies influence culture through political advocacy, advertising, and a variety of other efforts such as product placements in media. There is growing reputational risk and/or pressure on companies via social media and the press to source sustainable inputs, as well as increased scrutiny of supply chains for violations of human rights.

Social influences are often more significant for industries that sell to consumers rather than to other businesses. In concert with technological and economic influences, the global beauty industry (comprising skin care, color cosmetics, hair care, fragrances, and select personal care product categories) has grown briskly since 2010, particularly premium products. High-quality cameras on mobile devices and social media have increased the salience of people's physical appearance and created efficient marketing channels for beauty companies that utilize third-party influencers and video tutorials.

Technological Influences

Technological changes can create new or improved products or render existing products obsolete, radically changing industries and companies. Based on the pioneering work of Clayton Christensen, we distinguish between two kinds of technological changes: sustaining and disruptive innovations.

Sustaining innovations are improvements in product or service performance and the addition of marginal features, but without a fundamental change in functionality or operation. These innovations are often led by industry incumbents making improvements to better serve existing or adjacent customers' needs. Successive improvements to cable television—including digital transmission, improved video quality, premium programming, DVRs, and additional channels—represent sustaining innovations. Disruptive innovation is a change that creates a new market or enters an existing one but with a very different value proposition that may not immediately appeal to existing customers who are happy with the current product. Cable television was disrupted by streaming video delivered over the internet. At first a rudimentary offering that appealed to people who did not have cable subscriptions, it eventually grew to overtake cable in the number of subscribers.

Disruptive innovation is most likely to come from new entrants, because they do not have an existing profitable business to protect. The difficulty faced by incumbents with disruptive innovation is termed the "innovator's dilemma": they can either invest in the disruptive innovation themselves, speeding the decline of their existing business (and potentially harming employees) but not losing market share, or ignore the innovation and lose market share while continuing to generate strong profits in the near to medium term.

Legal Influences

Legal influences on an industry include changes in laws and regulations from courts and policymakers that alter an industry's business practices or economic outcomes. These influences, which are in addition to the fiscal and monetary policy changes discussed earlier under political influences, are often more subtle changes to the "rules of the road" governing industry affairs. Companies look to shape these influences through policy advocacy and in legal proceedings of their own.

Examples of industries strongly affected by legal influences include tobacco and cannabis. Jurisdictions have wide-ranging laws and regulations on the manufacturing, distribution, marketing, sale, and use of tobacco products to discourage consumption and the associated adverse health consequences, including secondhand smoke. Laws cover advertising prohibitions, graphic images on packaging, minimum purchase age, bans on smoking in public places, and limitations on nicotine content and flavorings. These laws are in addition to the generally high taxes levied on tobacco, which encourage non-taxable counterfeit activity that governments must police. Adoption of or changes in these laws in various jurisdictions significantly affect tobacco industry sales. In response to these influences, as well as broader social influences that have resulted in declining cigarette sale volumes, the tobacco industry is primarily focused on developing so-called reduced risk products that may pose fewer health risks and therefore create a more favorable legal environment.

The cannabis industry is in an evolving legal environment, where certain jurisdictions have legalized its production, sale, and use (Canada), while some have varying laws at the sub-sovereign level (the USA) and most others still completely prohibit cannabis (most of Asia).

Environmental Influences

Finally, environmental influences are often closely associated with legal influences and include risks and opportunities related to the transition to a lower-carbon economy, laws and practices concerning waste and land use, and environmental protections.

Transition risks and opportunities are important influences for sectors such as energy, airlines, utilities, and autos, as they are involved in the production or consumption of fossil fuels, which are responsible for carbon emissions. Due to changes in regulations, taxes, consumer preferences, or voluntary actions, companies in these sectors must evolve their business operations or face share losses to competing solutions with lower-carbon emissions.

QUESTION SET

1. In 2009, Amazon.com started Amazon Publishing, extending its business into publishing books in addition to book retailing. From the perspective of a competing publisher, this is an example of:

   A. decreasing the threat of substitutes.

   B. decreasing the bargaining power of suppliers.

   C. increasing the bargaining power of customers.

   Solution:

   C is correct. Amazon Publishing is an example of backward-integration, as it could now produce and sell its own published content as well as other publishers' content. The creation of Amazon Publishing increased the bargaining power of Amazon as a customer and increased the level of competitive rivalry in the publishing industry. Amazon Publishing would also increase the threat of substitutes and the bargaining power of suppliers, as authors and end users could work directly with Amazon.

2. Proposed government regulations for the restaurant industry would introduce new compliance requirements for all new and existing facilities to reduce the environmental impact of take-out and delivery options. These regulations would apply only to restaurants with a physical address and

would not apply to mobile restaurants (i.e., food trucks). For an existing restaurant with several physical locations, these regulations would:

- **A.** reduce competitive rivalry by increasing the barriers to entry.
- **B.** increase competitive rivalry by increasing the threat of substitutes.
- **C.** both reduce and increase competitive rivalry.

Solution:

C is correct. The proposed regulatory changes would have a mixed impact on the level of competitive rivalry in the restaurant industry. These environmental regulations would increase the entry barriers by increasing the costs of setting up a new restaurant, while also increasing the threat of substitutes by making it easier to open mobile restaurants, which do not have to comply with these regulations.

Please refer to the Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd. in the previous lesson to answer the following questions.

3. To compete with online retailers, Warehouse Club Inc. is considering installing a sophisticated image recognition system in its stores to track and analyze consumer behavior in order to identify clothing items that were tried on but not purchased. For Warehouse, this change would be a ____________________ (*sustaining* or *disruptive*) innovation.

Solution:

For Warehouse, this change would be a *sustaining* innovation. Sustaining improvements focus on adding marginal features without a fundamental change in functionality or operation. For example, the proposed system would provide incremental customer and product knowledge but would not represent a fundamental change in Warehouse's go-to-market strategy. Disruptive innovation is a change that creates a new market or enters an existing one with a different value proposition.

4. Identify the following statement as true or false. Justify your answer.

The CEO of Iliso Marketplace Ltd. expects revenue growth to be in line with US GDP growth for the next two fiscal years. As a result, the CEO approves an extension to an existing aggressive sales promotion and discount strategy that has been in place for the past 18 months. This extension will increase the competitive rivalry in the industry.

Solution:

False. The sales promotion and discount strategy is already in place within this mature industry, thus demonstrating a history of price competition among the competitors.

5. A PESTLE analysis for Iliso Marketplace Ltd. would *most likely* identify which of the following items?

- **A.** A competitor is acquiring a fleet of delivery vehicles instead of using third-party parcel carriers.
- **B.** The increasing political discussion around instituting a tax on parcel deliveries of less than 2 kilograms

C. A competitor is launching an augmented reality application to allow tech-savvy consumers to view home décor and furniture products in real time.

Solution:

B is correct. The political discussion around potential applications of a tax is an example of a political trend changing over time. This theme could potentially affect Iliso Marketplace Ltd. A (fleet of delivery vehicles) and C (augmented reality) are examples of competitive actions that could affect the level of competitive rivalry in the industry.

6 COMPETITIVE POSITIONING

☐ evaluate the competitive strategy and position of a company

All companies have a competitive strategy, whether intentional or not. An intentional strategy results from company-wide planning, performance measurement, and feedback loops to sharpen the strategy. An unintentional strategy results from various teams within a firm pursuing their own incentives, doing whatever they did in prior years, or following industry or professional norms. An unintentional strategy is rarely the best and often exacerbates communication and coordination problems present at every company, although the approach might perform well in areas such as discovery-oriented research. Munos (2009) found evidence of diseconomies of scale in pharmaceutical R&D, with most new drugs discovered by small companies operating in a relatively "unmanaged" way.

Ultimately, an effective competitive strategy is evidenced by a company's track record of value added for its stakeholders, such as economic profits. Its effectiveness can be judged only in hindsight. To evaluate a competitive strategy on a forward-looking basis, an analyst should assess the strategy along three dimensions:

- Does the strategy create a defense against the five industry forces?
- Does the strategy benefit from, or is at least not at odds with, the expected external industry influences identified in the PESTLE analysis?
- Does the company have the resources and capabilities to execute the strategy?

Clearly, the analysis and the answers to these questions are company and industry specific (CFA Institute has published a helpful industry-by-industry reference titled *Sector Analysis: A Framework for Investors* with examples). We can, however, identify three well-known competitive strategies that have worked in a variety of industries, as shown in Exhibit 6: cost leadership, differentiation, and focus. These three generic strategies are based on Michael Porter's research on competition. While companies do tend to have specific variations, generic strategies explain a large percentage of strategies in practice—similar to how many companies' business models are well described using conventional models. A poor situation is for a company to be "stuck in the middle"—not a cost leader, not differentiated, not focused.

Exhibit 6: Generic Competitive Strategies

	Cost Leadership	Differentiation	Focus
Means of executing strategy	▪ Economies of scale from fixed costs ▪ Favorable access to raw materials ▪ Culture of strict cost control ▪ Aggressive pricing to gain high volume ▪ Low-cost distribution ▪ Economies of scope	▪ Investments in advertising, brand, customer service, proprietary distribution channels ▪ Protection using trademarks, copyright, patents ▪ Superior quality, unique features ▪ Culture of strong customer experience ▪ Premium pricing ▪ Integration of services, software, and hardware	▪ Proximity to customers and strong understanding of their needs ▪ May incorporate elements of strategy from both cost leadership and differentiation, but focused on particular group
Which of the Five Forces it defends against (why it works)	▪ Threat of new entrants: Capital requirements and scale advantages deter entrants ▪ Bargaining power of customers: Customers can only bring prices down to the costs of the marginal producer, leaving margin for the cost leaders ▪ Industry rivalry: Rivals may not be able to compete on price with cost leaders	▪ Threat of new entrants and of substitutions: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable to unwilling to comparison shop or switch ▪ Bargaining power of suppliers: The company may have the ability to pass along price increases to customers and/or margin to absorb cost increases	▪ Threat of new entrants and of substitutes: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable or unwilling to comparison shop or switch
Industry appropriateness	▪ Capital intensive ▪ Price-conscious customers ▪ Customers do not value or notice product differences ▪ Minimal innovation in industry	▪ Price is not foremost concern for customers ▪ Customers value distinctiveness ▪ Innovation in industry, with products varying in features and forms	▪ Difficult (or uneconomical) to serve customer group, product, or geography for other players
Risks to the strategy	▪ Cost inflation, loss of discipline ▪ Technological change that results in loss of cost leadership or market share ▪ Desire for premiumization among customers	▪ Imitation by competitors ▪ Buyers become sophisticated, no longer demand level of service ▪ Pricing premium becomes too high for customers to bear ▪ May preclude high market share, as customers value exclusivity	▪ Larger competitors outcompete on price ▪ The differences in demand between the between the narrow group and industry as a whole narrow ▪ Buyers become sophisticated, no longer demand level of service

QUESTION SET

1. Manitou Resorts is a 25-room boutique hotel located in a rural mountain area approximately three hours from a major urban center. It provides accommodations, high-end dining options, and an on-site health and wellness spa. Manitou's primary customers are young professional couples

in the middle-to-upper-income brackets. The CEO is contemplating several options for expansion, as the resort is frequently at full capacity. Which of the following initiatives would be consistent with Manitou's current competitive strategy?

A. The development of 10 stand-alone lakefront cottages to accommodate families of up to six people, providing access to family-related activities such as pools, water sports, and evening entertainment

B. The addition of conference room facilities to accommodate corporate events that could be priced at a 10%–15% discount to a 150-room franchise hotel approximately 30 minutes from Manitou

C. Developing a management and reservation system to identify recommendations to referral partners such as golf courses, taxi/shuttle services, or other guest experiences to increase fee revenue

Solution:

C is correct. Manitou Resorts' current competitive strategy is a focus strategy targeting a specific customer segment: young professionals in middle-to-upper-income brackets located in a nearby city and focused on wellness and relaxation. The other options represent an expansion into new customer segments (family vacations and corporate events), which could be inconsistent with its current customer base. A represents a cost leadership competitive strategy, while B represents a differentiation strategy.

2. Warehouse Club Inc. is considering launching a "private label" brand of non-perishable food products. Based on its current competitive strategy, which of the following is the *most appropriate* rationale to support?

A. Customers do not notice product differences.

B. Customers value new product features and forms.

C. Products can be segmented into regular and premium brands.

Solution:

A is correct. Warehouse Club Inc.'s current competitive strategy is cost leadership. In a cost leadership strategy, industry appropriateness is determined by capital intensity, price-conscious customers, customers not valuing or noticing product differences, and minimal innovation in the industry. B represents an example of a differentiation strategy. C represents an example of a focus strategy.

3. To defend against the bargaining power of suppliers, customers may be unable or unwilling to comparison shop or switch in which of the following competitive strategies (select all that apply):

___ Cost leadership

___ Differentiation

___ Focus

Solution:

Both differentiation and focus. That "customers may be unable or unwilling to comparison shop or switch" is a common feature of both differentiation and focus strategies. In a cost leadership strategy, "customers can only bring prices down to the costs of the marginal producer, leaving margin for the cost leaders."

4. A risk to a differentiation competitive strategy is:

   A. limited market share due to exclusivity.

   B. larger competitors outcompete on price.

   C. technological change that results in loss of market share.

   Solution:

   A is correct. Since a differentiation strategy pursues a unique product or service offering, it creates a risk that the strategy may preclude high market share, as customers value exclusivity. B is a risk to a focus strategy where larger competitors outcompete on price. C is a risk to a cost leadership strategy where the technological change results in cost leadership or market share loss.

5. A large online retailer in the USA has begun to offer a "price-matching" guarantee on its website to match any competitor's online price for an identical product, including Iliso Marketplace. Recommend and justify whether Iliso should launch a similar guarantee to protect its market share.

   Solution:

   No, Iliso should not launch a similar guarantee. A price-matching guarantee is a strategy pursued by companies with a cost leadership approach by using aggressive pricing to gain volume. Iliso Marketplace Ltd. operates with a differentiation strategy that permits premium pricing if aligned with a strong customer experience. If Iliso maintained its customer experience offering and began to reduce its revenue to match cost leaders, the company would likely decrease its overall profitability.

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PRACTICE PROBLEMS

1. An alternative method of grouping companies by geography is *least likely* to be completed using:

   A. location of head office.

   B. geographic composition of revenue.

   C. primary listing of its equity securities.

2. A market consists of three firms with market shares of 50%, 30%, and 20%. The Herfindahl-Hirschman Index (HHI) is *closest* to:

   A. 0.38.

   B. 2,500.

   C. 3,800.

3. Lower industry concentration is usually associated with a high degree of competitive intensity unless the industry is *most likely*:

   A. global.

   B. service-oriented.

   C. one with low product differentiation.

4. Increased environmental regulations on automotive manufacturers will *most likely* have which of the following effects on the threat of new entrants?

   A. An increase

   B. A decrease

   C. No change

5. The CEO of a large law firm is concerned about a new mobile application that uses algorithms to auto-complete legal forms and questions in discovery and provides recommendations for small-claims matters at a much lower cost than a traditional law firm charges. As a result, the CEO is contemplating whether the firm should develop and launch its own branded application, which would affect its short-term profitability but maintain its existing customer base, or give up market share to focus on more complex claims with higher profitability. This scenario is an example of:

   A. defensiveness.

   B. innovator's dilemma.

   C. sustaining innovation.

6. PESTLE analysis is a framework for identifying:

   A. industry themes.

   B. the level of industry concentration.

C. determinants of industry profitability.

7. The means of executing cost leadership competitive strategies *do not include* which of the following?

   A. proximity to customers

   B. favorable access to raw materials

   C. economies of scale from fixed costs

8. Which of the following is *most likely* a risk of executing a differentiation competitive strategy?

   A. pricing premiums become too high

   B. larger competitors outcompete on price

   C. a desire for premium-ization among customers

SOLUTIONS

1. B is correct. Classification by country is typically by the country where the issuer is incorporated, the country of the primary listing of its equity securities, the location of its headquarters, or market perception. Note that classification by the geographic composition of revenue is generally *not* the approach taken, though this aspect may be the foremost concern for an analyst.

2. C is correct. The Herfindahl-Hirschman Index (HHI) is calculated as the sum of the squares of competitor market shares. In this example, the HHI is calculated as ($50^2 + 30^2 + 20^2 = 3,800$). A incorrectly calculates the HHI using the percentage market shares ($0.5^2 + 0.3^2 + 0.2^2 = 0.38$). B incorrectly calculates the HHI using only the market leader concentration ($50^2 = 2,500$).

3. B is correct. Lower industry concentration, defined as many small competitors in the market, is usually associated with a high degree of competitive intensity unless the industry is service-oriented, is local in nature, or has high product differentiation.

4. B is correct. Increased environmental regulations on automotive manufacturers represent an additional cost for compliance when building a new manufacturing facility and increase the barriers to entry into the industry.

5. B is correct. The described mobile application represents a fundamental change in the business model of a law firm. The application is a disruptive, not sustaining, innovation, as it brings new entrants into the market but with a very different value proposition. The CEO's decision represents the "innovator's dilemma": the firm can either invest in the disruptive innovation, speeding the decline of its existing business but not losing market share, or ignore the innovation and lose market share while continuing to generate strong profits in the near term.

6. A is correct. PESTLE analysis is a framework for identifying "themes" or "narratives" that investors may take a perspective on and desire exposure to. B refers to the Herfindahl-Hirschman Index. C refers to Porter's Five Forces.

7. A is correct. Proximity to customers is a means of executing a focus strategy. The means of executing a cost leadership strategy include economies of scale from fixed costs, favorable access to raw materials, a culture of strict cost control, aggressive pricing to gain high volume, low-cost distribution, and economies of scope.

8. A is correct. The risks of a differentiation strategy include imitation by competitors, buyers becoming sophisticated and no longer demanding level of service, and pricing premiums becoming too high for customers to beat; such a strategy may also preclude high market share, as customers value exclusivity. B relates to a focus strategy, while C relates to a cost leadership strategy.